

Hardware Sensor Test Documentation: Accelerometer

Module: Accelerometer Sensor Testing

Functionality: Screen Orientation & Rotation

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_ACC_001	Verify UI rotation from Portrait to Landscape mode.	Auto-rotate feature is enabled in Quick Settings. App supporting multi-orientation is open.	1. Hold the physical device vertically (Portrait). 2. Rotate the device 90 degrees horizontally to the left or right.	The sensor must detect the change along the X/Y axes, and the UI layout must instantly and smoothly switch to Landscape mode.	Pass
TC_ACC_002	Verify UI rotation behavior when Auto-rotate is disabled.	Auto-rotate feature is toggled OFF in Settings.	1. Open a gallery or browser app. 2. Rotate the physical device from Portrait to Landscape mode.	The Accelerometer may register the raw movement, but the system must block the UI layout transition, keeping it locked in Portrait.	Pass
TC_ACC_003	Verify 180-degree reverse portrait orientation block.	Device is on the home screen with auto-rotate ON.	1. Turn the physical device completely upside down (180 degrees reverse portrait).	Unless explicitly supported by the launcher app, the system should ignore or block 180-degree inverted layout rotation.	Pass

Functionality: Gesture & System Interactions

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_ACC_004	Verify “Raise to Wake” system gesture detection.	“Raise to Wake” gesture is toggled ON in System Settings.	1. Place the device flat on a stable surface. 2. Wait 10 seconds for the screen to timeout/lock. 3. Lift the device naturally toward face level.	The sensor must detect sudden linear acceleration change on the Z-axis and immediately wake up the lock screen without power button input.	Pass
TC_ACC_005	Verify “Flip to Mute” incoming call gesture.	“Flip to Mute” settings are activated on the device.	1. Trigger an incoming call from a secondary phone. 2. Turn the physical device face down on a flat desk.	The accelerometer must register the sharp pitch change, and the system must immediately mute the ringtone audio/vibration.	Pass
TC_ACC_006	Verify shake gesture handling (e.g., Shake to clear/flashlight).	An app or system tool utilizing shake gestures is enabled.	1. Hold the device firmly. 2. Perform a sudden, repetitive left-and-right shaking motion.	The raw G-force thresholds must trigger the designated action (e.g., undo typing or turning on flashlight) successfully.	Pass

Functionality: Hardware Calibration & Edge Cases

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_ACC_007	Verify raw sensor values on a perfectly flat surface.	Factory Diagnostic Tool or Sensor Test App is open. Device is on a calibrated flat surface.	1. Leave the device completely static. 2. Observe raw output values for X, Y, and Z axes.	Raw values should read approximately: $X \approx 0 \text{ m/s}^2$, $Y \approx 0 \text{ m/s}^2$, $Z \approx 9.8 \text{ m/s}^2$ (reflecting earth's gravity standard).	Pass
TC_ACC_008	Verify sensor responsiveness during sudden rapid vibrations.	Device is subjected to sudden erratic movements (e.g., driving on rough roads or running).	1. Run a real-time sensor logging script. 2. Vibrate or drop the device slightly onto a cushioned surface.	The sensor sampling rate must be constant without drops, spikes to infinite values, or internal thread deadlocks.	Pass
TC_ACC_009	Verify sensor power state during System Sleep / Deep Sleep mode.	Device goes to idle state with screen OFF for a prolonged duration.	1. Monitor system sensor power logs via ADB tool. 2. Let the device enter sleep mode.	The system must place the accelerometer into a low-power polling state to save battery, waking it instantly upon physical movement.	Pass